

Unit 5

LU Decomposition

Unit Goals

- Be able to explain the relationship between LU decomposition and Gaussian elimination.
- Be able to LU-decompose an arbitrary matrix, $A = LU$.
- Be able to explain how row exchanges lead to an LU decomposition including a permutation matrix, $PA = LU$.
- Be able to use the result of LU decomposition to solve linear systems efficiently.
- Be able solve real-world problems with linear models using LU decomposition.

Unit 5.1

What is LU decomposition?

Advantages of the matrix inverse.

- The matrix inverse “stores” the result of Gaussian elimination, allowing solutions to be found with new right-hand sides easily and efficiently.
 - Gaussian elimination for a system of n equations and n unknowns requires about $\left(\frac{2}{3}\right)n^3$ total arithmetic operations; **it is an $O(n^3)$ problem.**
 - Matrix multiplication of an $n \times n$ matrix and an $n \times 1$ vector requires about $2n^2$ total arithmetic operations; **it is an $O(n^2)$ problem.**
- The matrix inverse provides a framework in symbolic matrix algebra to represent solving a linear system, which can be used to gain insight and to develop new approaches to problems.

Limitations of the matrix inverse

- The matrix inverse does not exist for non-square matrices, nor for all square matrices.
- A system of equations in more equations than unknowns may have a unique solution.
- Systems of equations in fewer equations than unknowns may have a solution for RHS, just that none is unique.

Limitations of the matrix inverse

- The matrix inverse doesn't help us with systems that have unique solutions for some RHS.
- The matrix inverse doesn't help us find the general solution to a system with **infinite solutions**.
- The matrix inverse doesn't help us determine which right-hand sides lead to a consistent system.
- **All these problems we were able to handle with Gaussian Elimination.**

The forward steps of Gaussian elimination are the most important.

- Forward elimination leads a checkpoint on existence and uniqueness of solutions.
- Forward elimination is the most computationally costly part of the problem.
 - Forward elimination for a system of m equations and n unknowns requires about $\left(\frac{2}{3}\right)m^2n$ total arithmetic operations; it is an $O(m^2n)$ problem.
 - Back substitution for a system of m equations and n unknowns requires only about mn total arithmetic operations; is an $O(mn)$ problem.

Goal of the LU decomposition

- Conceptually store everything about the forward elimination steps of Gaussian elimination.
- Similar reduction of work for multiple right-hand-sides as the matrix inverse.
- Applicable to both square and non-square systems.
- Can be used to derive the unique solution if there is one.
- Can be used to derive the general solution when multiple solutions exist.
- Can be used to determine whether a particular right-hand side leads to a consistent or inconsistent system.

A summary of LU decomposition matrices.

We will use the following notation in our discussion

- A : The original matrix.
 - In general this will be the coefficient matrix of a linear system, not the augmented matrix.
- U : The echelon (upper triangular) form of A .
 - This is the matrix found after only the forward elimination step.
- L : A lower triangular matrix.
 - This matrix stores the information about forward elimination.
- P : A permutation matrix.
 - If row exchanges were used, this matrix stores them.

Two forms of LU decomposition.

- When row exchanges are not needed, the LU decomposition has the form:

$$A = LU$$

- The original matrix is expressed as a product of a lower triangular and an upper triangular matrix.
- When row exchanges are used, the LU decomposition has the form:

$$PA = LU$$

- The original matrix first has its rows rearranged, then the result is expressed as a product of a lower triangular and an upper triangular matrix.

Benefit of the LU decomposition

- There are many matrix decompositions used in linear algebra.
- They have the common feature of expressing a matrix as a product of matrices that have a special form.
- These special forms are easier to work with!

- Consider:

$$A\vec{x} = \vec{b}, \quad A = LU \rightarrow LU\vec{x} = \vec{b} \rightarrow L\vec{c} = \vec{b}, \quad U\vec{x} = \vec{c}$$

- One system is replaced by two, but both are of a special form:
 - Step 1: Solve the system $L\vec{c} = \vec{b}$, to find $\vec{c}$. Easy: L is lower triangular.
 - Step 2: Solve the system $U\vec{x} = \vec{c}$, to find $\vec{x}$. Easy: U is upper triangular.

Unit 5.2

Deriving the LU decomposition.

Gaussian Elimination is reversible.

- Every elementary row operation has a corresponding inverse operation that “undoes” the effect:
 - Addition of a scalar multiple is reversed by subtracting the same scalar multiple.
 - Scalar multiplication is reversed by multiplying by the inverse of the scalar.
 - Row exchange is reversed by repeating the same exchange.

Original Operation

$$R_i + kR_j \rightarrow R_i$$

$$kR_i \rightarrow R_i$$

$$R_i \leftrightarrow R_j$$

Inverse Operation

$$R_i - kR_j \rightarrow R_i$$

$$\frac{1}{k}R_i \rightarrow R_i$$

$$R_i \leftrightarrow R_j$$

Elementary matrices are non-singular.

- Every elementary row operation has a corresponding inverse operation that “undoes” the effect:

$$A \xrightarrow{R_f} A' \xrightarrow{R_r} A$$

- We can define matrices for both for the forward and inverse operations:

$$A' = E_f A, \quad A = E_r A'$$

- The elementary matrix for the inverse operation is the inverse of the elementary matrix for the forward operation!

$$\begin{aligned} A &= E_r E_f A, \forall A \rightarrow E_r E_f = I \\ A' &= E_f E_r A', \forall A \rightarrow E_f E_r = I \\ E_r &= (E_f)^{-1} \end{aligned}$$

The inverse of the forward elimination matrix reverses elimination.

- Recall that forward elimination can be represented by a product of elementary matrices:

$$A \xrightarrow{R_{f1}} A_{f1} \xrightarrow{R_{f2}} A_{f2} \xrightarrow{R_{f3}} \cdots \xrightarrow{R_{fr-1}} A_{fr-1} \xrightarrow{R_{fr}} A_{fr} = U$$

$$A \rightarrow E_{f1}A \rightarrow E_{f2}(E_{f1}A) \rightarrow \cdots \rightarrow E_{fr} \left(E_{fr-1} \left(\cdots E_{f2}(E_{f1}A) \right) \right) = U$$

$$FA = U, \quad F = \left(\prod_{i=0}^{r-1} E_{fr-i} \right)$$

The inverse of the forward elimination matrix reverses elimination.

- We can apply the same idea, in reverse, to obtain a matrix that reverses elimination:

$$U = A_{fr} \xrightarrow{R_{r1}} A_{fr-1} \xrightarrow{R_{r2}} A_{fr-2} \xrightarrow{R_{r3}} \cdots \xrightarrow{R_{rr-1}} A_{f1} \xrightarrow{R_{rr}} A$$

$$U \rightarrow E_{r1}U \rightarrow E_{r2}(E_{r1}U) \rightarrow \cdots \rightarrow E_{rr} \left(E_{rr-1}(\cdots E_{r2}(E_{r1}U)) \right) = A$$

$$RU = A, \quad R = \left(\prod_{i=0}^{r-1} E_{rr-i} \right)$$

The inverse of the forward elimination matrix reverses elimination.

$$FA = U, \quad F = \left(\prod_{i=0}^{r-1} E_{fr-i} \right) \quad RU = A, \quad R = \left(\prod_{i=0}^{r-1} E_{rr-i} \right)$$

- The elementary matrices for the forward and reverse steps are related: the $(i+1)^{\text{th}}$ reverse step is the inverse of the $(r-i)^{\text{th}}$ forward step (and vice versa). Thus:

$$E_{ri+1} = (E_{fr-i})^{-1} \quad E_{fi+1} = (E_{rr-i})^{-1}$$
$$R = \left(\prod_{i=0}^{r-1} E_{rr-i} \right) = \left(\prod_{i=1}^r (E_{fi})^{-1} \right) = \left(\prod_{i=0}^{r-1} E_{fr-i} \right)^{-1} = F^{-1}$$

The inverse of the forward elimination matrix reverses elimination.

- The inverse elimination matrix is the product of the inverse of the forward elimination matrices, with the product order reversed:

$$FA = U, \quad F = \left(\prod_{i=0}^{r-1} E_{fr-i} \right) \quad A = F^{-1}U, \quad F^{-1} = \left(\prod_{i=1}^r (E_{fi})^{-1} \right)$$

- What do F and F^{-1} look like, and can we compute them easily?

Forward elimination only needs one or two types of row operation.

- When we follow the forward elimination algorithm strictly, we never scale rows; this is part of back substitution.
- In many (most) cases, row exchange is not needed either!
- **Primary elimination operation:** $R_k - \left(a_{kj}/a_{ij}\right)R_i \rightarrow R_k$
- **Occasional row exchange:** $R_i \leftrightarrow R_j$
- Consider the elementary matrices for these.

Elementary elimination matrices.

- An example:

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_3 - 3R_1 \rightarrow R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} = E$$

- General form: $R_k - (a_{kp}/a_{lp})R_l \rightarrow R_k$ (l is the pivot row and p is the pivot column)

$$E = [e_{ij}], e_{ij} = \begin{cases} 1, & \text{if } i = j \\ -(a_{kp}/a_{lp}), & \text{if } i = k, j = l \\ 0, & \text{otherwise} \end{cases}$$

- An identity matrix with one non-zero off-diagonal element.
- Scaling factor located at row to be replaced and column matching the row used for elimination – always below the diagonal for forward elimination ($k > l$).

The product of (forward) elimination matrices is lower triangular.

- The elementary matrices of forward elimination are lower triangular, with ones on the diagonal.
- The product of any such matrices is lower triangular with ones on the diagonal.

- **An example:** $A \xrightarrow{(1) R_2 - 2R_1 \rightarrow R_2} \xrightarrow{(2) R_3 - 3R_1 \rightarrow R_3} \xrightarrow{(3) R_3 + R_2 \rightarrow R_3} U$

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}, E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$F = E_3 E_2 E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -5 & 1 & 1 \end{bmatrix}$$

The product of the inverse elimination matrices is lower triangular.

- The inverse of the forward elimination steps are just the same steps applied in reverse order with a sign change.

$$\text{Forward: } R_k - \left(a_{kp}/a_{lp}\right)R_l \rightarrow R_k \quad \text{Inverse: } R_k + \left(a_{kp}/a_{lp}\right)R_l \rightarrow R_k$$

- The same properties must thus apply.

- **An example:** $U \xrightarrow{(-3) R_3 - R_2 \rightarrow R_3} \xrightarrow{(-2) R_3 + 3R_1 \rightarrow R_3} \xrightarrow{(-1) R_2 + 2R_1 \rightarrow R_2} A$

$$E_{-1} = \begin{bmatrix} 1 & 0 & 0 \\ +2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} E_{-2} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ +3 & 0 & 1 \end{bmatrix}, E_{-3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$R = F^{-1} = E_{-1}E_{-2}E_{-3} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 1 & 1 \end{bmatrix}$$

Without row exchange, forward elimination gives L and U.

- We can write forward elimination and its inverse as a matrix product.

$$FA = U, \quad F = \left(\prod_{i=0}^{r-1} E_{fr-i} \right) \quad A = F^{-1}U, \quad F^{-1} = \left(\prod_{i=1}^r (E_{fi})^{-1} \right)$$

- If we do not use any row exchanges (or scale any rows), F and F^{-1} are both lower triangular with ones on the diagonal.
- F^{-1} is L :

$$L \equiv F^{-1} \rightarrow A = LU, \quad U = L^{-1}A$$

Elementary row exchange matrices.

- An example

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = E$$

- General form:

$$R_k \leftrightarrow R_l$$
$$E = [e_{ij}], e_{ij} = \begin{cases} 1, & \text{if } i = j, & i, j \neq k, l \\ 1, & \text{if } i = k, j = l \\ 1, & \text{if } i = l, j = k \\ 0, & \text{otherwise} \end{cases}$$

- An identity matrix with two rows swapped.

The product of row exchange matrices is a permutation matrix.

- Each multiplication corresponds a row swap; if we start with the identity matrix, we get an identity matrix with scrambled rows!
- There will be exactly one “1” in every row and in every column, and all other entries will be “0”.

- **An example:**

$$A \xrightarrow{(1) R_1 \leftrightarrow R_3} \xrightarrow{(2) R_2 \leftrightarrow R_3} A'$$

$$E_1 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$P = E_2 E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Isolating row exchanges lead to $PA=LU$.

- If we knew in advance where we would need to exchange rows, we could do so before starting elimination.

$$A \xrightarrow{\text{Row Exchanges}} A' \xrightarrow{\text{Elimination}} U$$

$$A' = PA$$

$$U = L^{-1}A' = L^{-1}PA$$

$$PA = LU$$

- P is the matrix that executes row exchanges.
- L^{-1} is the inverse of the matrix that executes the elimination steps.

Unit 5.3

Using the LU decomposition.

A simple way to generate L.

$$\begin{array}{ccc} A & \xrightarrow{\text{Row Exchanges}} & A' \xrightarrow{\text{Elimination}} U \\ & A' = PA, & U = L^{-1}A' \\ & PA = LU & \end{array}$$

- L is a lower triangular matrix that reverses the elimination steps, after we have appropriately arranged the rows:

$$L \equiv F^{-1} = \left(\prod_{i=1}^r (E_{fi})^{-1} \right)$$

A simple way to generate L.

$$L \equiv F^{-1} = \left(\prod_{i=1}^r (E_{fi})^{-1} \right)$$

- Each matrix in the product is the inverse of the corresponding forward elimination matrix.
- The reverse step involves the same operation with a different sign:

$$\text{Forward: } R_i - kR_j \rightarrow R_i \qquad \text{Reverse: } R_i + kR_j \rightarrow R_i$$

- We can directly write down these matrices as we perform elimination!

A simple way to generate L.

- We can directly write down these matrices as we perform elimination!

$$PA = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 1 \\ -1 & 2 & 1 \end{bmatrix} \xrightarrow[\text{(2): } R_3 + R_1 \rightarrow R_3]{\text{(1): } R_2 - 2R_1 \rightarrow R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -1 \\ 0 & 3 & 2 \end{bmatrix} \xrightarrow{\text{(3): } R_3 + 3R_2 \rightarrow R_3} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix} = U$$

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ +1 & 0 & 1 \end{bmatrix}, E_1 E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & +3 & 1 \end{bmatrix}$$

$$(E_1)^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ +2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, (E_2)^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}, (E_3)^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}$$

A simple way to generate L.

$$L \equiv F^{-1} = \left(\prod_{i=1}^r (E_{fi})^{-1} \right)$$

- A remarkable thing happens with **when** we take the product to find L:
- The product simply has all of the individual scaling terms in their appropriate position!

$$L = (E_1)^{-1}(E_2)^{-1}(E_3)^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ +2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ +2 & 1 & 0 \\ -1 & -3 & 1 \end{bmatrix}$$

- The same is not true of $L^{-1} = E_3 E_2 E_1$ due to the difference in order of multiplication.

A simple way to generate L.

Given an $m \times n$ coefficient matrix A :

- Create an $m \times m$ empty matrix that will become L .
- Place ones along the diagonal of L and zeros above it.
- Apply Gaussian Elimination (forward steps) converting $A \rightarrow U$.
- With each elimination step, note the row of the pivot: call this p .
- Fill in column p of L (below the diagonal) with the multiple of the pivot row that was *subtracted* from each row below it:

$$R_k - xR_p \rightarrow R_k \Rightarrow l_{kp} = x$$

- Take care not to forget the cases where $x = 0$.

A simple way to generate L.

- An example:

$$PA = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 1 \\ -1 & 2 & 1 \end{bmatrix} \xrightarrow[\text{(2): } R_3 + R_1 \rightarrow R_3]{\text{(1): } R_2 - 2R_1 \rightarrow R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -1 \\ 0 & 3 & 2 \end{bmatrix} \xrightarrow{\text{(3): } R_3 + 3R_2 \rightarrow R_3} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix} = U$$

$$\begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ & & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -3 & 1 \end{bmatrix} = L$$

$$PA = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 1 \\ -1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix} = LU$$

Be careful with row exchanges!

- Note: The direct approach to L works ONLY when there are no row exchanges applied during the elimination steps.
- If you do need row exchanges, they need to be applied first!
- “By-hand” approach (more work, but straightforward):
 1. Perform forward elimination to identify any necessary row exchanges.
 2. Apply these to the original matrix and repeat forward elimination to generate L .
- “Computational” approach (less work, but error-prone):
 - When we reach a row exchange, reconstruct the sequence of operations that would have been needed, and adjust L directly.

LU decomposition checkpoints.

- The U matrix is the end point of forward elimination.
- Checkpoints:
 - If U has no zero rows, then the system has a solution for every possible right-hand side.
 - If U has a pivot in every column (i.e. there are no free variables) then any solution must be unique.
 - For square matrices, these must both be true (for a non-singular system) or both be false (for a singular system).
 - For non-square matrices, these are independent checkpoints!

Solving systems with LU decomposition.

- The LU decomposition exists for EVERY matrix.
- It can be applied to ANY linear system.

• Consider: $A\vec{x} = \vec{b}, PA = LU$

$$LU\vec{x} = P\vec{b} \rightarrow \vec{b}' = P\vec{b}, \quad L\vec{c} = \vec{b}', \quad U\vec{x} = \vec{c}$$

1. Re-order RHS, if needed: $\vec{b}' = P\vec{b}$
2. Solve Lower Triangular system: $L\vec{c} = \vec{b}'$
3. Solve Upper Triangular system: $U\vec{x} = \vec{c}$

- If $\vec{c}$ has a non-zero entry where U has a zero row, no solution.
- If U has a free variables, $\vec{x}$ is not unique

Solving systems with LU decomposition.

$$A\vec{x} = \vec{b}, PA = LU$$

$$\vec{b}' = P\vec{b}$$

$$L\vec{c} = \vec{b}'$$

$$U\vec{x} = \vec{c}$$

- Example: Solve the system $A\vec{x} = \vec{b}$

$$\text{for } \vec{b} = [1 \quad -1 \quad 2]^T$$

given the LU decomposition of **A**, $PA = LU$

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -3 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix}$$

Solving systems with LU decomposition.

$$A\vec{x} = \vec{b}, PA = LU$$

$$\vec{b}' = P\vec{b}$$

$$L\vec{c} = \vec{b}'$$

$$U\vec{x} = \vec{c}$$

• Step 1: $\vec{b}' = P\vec{b} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$

• Step 2:

$$[L \quad \vec{b}'] = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 2 & 1 & 0 & 1 \\ -1 & -3 & 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & -3 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -8 \end{bmatrix} = [I \quad \vec{c}]$$

• Step 3:

$$[U \quad \vec{c}] = \begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & -1 & -1 & -3 \\ 0 & 0 & -1 & -8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 & -6 \\ 0 & -1 & 0 & 5 \\ 0 & 0 & 1 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -5 \\ 0 & 0 & 1 & 8 \end{bmatrix} = [I \quad \vec{x}]$$

Understanding LU decomposition

$$\begin{array}{ccccc} [A & \vec{b}] & \xrightarrow{\text{Row Exchanges}} & [A & \vec{b}'] & \xrightarrow{\text{Elimination}} & [U & \vec{c}] & \xrightarrow{\text{Back Substitution}} & [R & \vec{x}] \\ & \vec{b}' = P\vec{b} & & L\vec{c} = \vec{b}' & & U\vec{x} = \vec{c} \end{array}$$

Two steps give us the correct RHS to match U.

1. Re-order RHS, if needed: $\vec{b}' = P\vec{b}$
2. Solve Lower Triangular system: $L\vec{c} = \vec{b}'$
 - $\vec{c}$ can also be found by simply applying the steps of forward elimination to $\vec{b}$.

Then we solve with back substitution.

3. Solve Upper Triangular system: $U\vec{x} = \vec{c}$

Unit 5.3.1

Worked examples finding the LU decomposition.

Find the LU decomposition

- Find L and U such that $A=LU$, given:

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 4 & 6 & 4 \\ -1 & 0 & 0 & 1 \\ 3 & 3 & 4 & 2 \end{bmatrix}$$

Find the LU decomposition

- Perform Forward Elimination to get U and encode L:

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 4 & 6 & 4 \\ -1 & 0 & 0 & 1 \\ 3 & 3 & 4 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 4 & 2 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 4 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 4 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & 0 \\ & & 1 & 0 \\ & & & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & & 1 & 0 \\ 3 & & & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 1/2 & 1 & 0 \\ 3 & 0 & & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 1/2 & 1 & 0 \\ 3 & 0 & -1 & 1 \end{bmatrix}$$

Find the LU decomposition

- Verify:

$$LU = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 1/2 & 1 & 0 \\ 3 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 4 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & (2+2) & (2+4) & (2+2) \\ -1 & (-1+1) & (-1+2-1) & (-1+1+1) \\ 3 & 3 & (3+1) & (3-1) \end{bmatrix} =$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 4 & 6 & 4 \\ -1 & 0 & 0 & 1 \\ 3 & 3 & 4 & 2 \end{bmatrix} = A$$

Find the LU decomposition (with P)

- Find P, L and U such that $PA=LU$, given:

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 2 & 2 & 4 & 4 \\ 1 & 1 & 2 & 3 \\ 2 & 0 & 2 & 0 \end{bmatrix}$$

Find the LU decomposition

- Perform Forward Elimination to identify row swaps, and generate P:

$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 2 & 2 & 4 & 4 \\ 1 & 1 & 2 & 3 \\ 2 & 0 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 2 & 3 \\ 2 & 0 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & -2 & -2 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -2 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} = P$$

Find the LU decomposition

- Perform Forward Elimination to get U and encode L:

$$PA = \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 2 & 0 & 2 & 0 \\ 1 & 1 & 2 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & -2 & -2 & -4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix} = U$$

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & 0 \\ & & 1 & 0 \\ & & & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & 0 \\ & & 1 & 0 \\ 1/2 & & & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & 0 \\ & & 1 & 0 \\ 1/2 & -2 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & 0 \\ & & 1 & 0 \\ 1/2 & 0 & 0 & 1 \end{bmatrix}$$

Find the LU decomposition

- Verify:

$$\begin{aligned}
 LU &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 1/2 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 2 & (2-2) & (4-2) & (4-2-2) \\ 1 & 1 & 2 & (2+1) \end{bmatrix} = \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 2 & 0 & 2 & 0 \\ 1 & 1 & 2 & 3 \end{bmatrix} = PA
 \end{aligned}$$

LU decomposition of a linear system

- Express the given linear system as a pair of triangular systems, $L\vec{c} = \vec{b}$ and $U\vec{x} = \vec{c}$.

$$\begin{array}{cccccccl} 2x_1 & + & 3x_2 & + & 4x_3 & + & 4x_4 & = & 2 \\ 2x_1 & + & x_2 & + & 4x_3 & + & 3x_4 & = & 5 \\ 4x_1 & + & 7x_2 & + & 8x_3 & + & 6x_4 & = & 3 \end{array}$$

LU decomposition of a linear system

- Express in matrix form:

$$\begin{array}{cccccccl} 2x_1 & + & 3x_2 & + & 4x_3 & + & 4x_4 & = & 2 \\ 2x_1 & + & x_2 & + & 4x_3 & + & 3x_4 & = & 5 \\ 4x_1 & + & 7x_2 & + & 8x_3 & + & 6x_4 & = & 3 \end{array}$$

$$A\vec{x} = \vec{b}, \quad A = \begin{bmatrix} 2 & 3 & 4 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 7 & 8 & 6 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{b} = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}$$

Find the LU decomposition

- Perform Forward Elimination on A to get U and encode L:

$$A = \begin{bmatrix} 2 & 3 & 4 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 7 & 8 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 3 & 4 & 4 \\ 0 & -2 & 0 & -1 \\ 0 & 1 & 0 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 3 & 4 & 4 \\ 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & -5/2 \end{bmatrix} = U$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ & & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -1/2 & 1 \end{bmatrix}$$

Find the LU decomposition

- Define two triangular systems:

$$A\vec{x} = \vec{b}, A = LU \rightarrow L(U\vec{x}) = \vec{b} \rightarrow L\vec{c} = \vec{b}, U\vec{x} = \vec{c}$$

$$L\vec{c} = \vec{b}, \quad L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -1/2 & 1 \end{bmatrix}, \vec{c} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}$$

$$U\vec{x} = \vec{c}, \quad U = \begin{bmatrix} 2 & 3 & 4 & 4 \\ 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & -5/2 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

Unit 5.3.2

**Worked examples using the LU
decomposition.**

Solve, given L and U.

- Solve the linear system $A\vec{x} = \vec{b}$, given:

$$A = LU,$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -1/2 & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & 3 & 4 & 4 \\ 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & -5/2 \end{bmatrix}, \vec{b} = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}$$

Solve, given L and U.

- Set up two triangular systems:

$$A\vec{x} = \vec{b}, A = LU \rightarrow L(U\vec{x}) = \vec{b} \rightarrow L\vec{c} = \vec{b}, U\vec{x} = \vec{c}$$

$$L\vec{c} = \vec{b}, \quad L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -1/2 & 1 \end{bmatrix}, \vec{c} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}$$

$$U\vec{x} = \vec{c}, \quad U = \begin{bmatrix} 2 & 3 & 4 & 4 \\ 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & -5/2 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

Solve, given L and U.

- Solve $L\vec{c} = \vec{b}$:

$$L\vec{c} = \vec{b}, L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -1/2 & 1 \end{bmatrix}, \vec{c} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}$$

$$M = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 1 & 1 & 0 & 5 \\ 2 & -1/2 & 1 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & -1/2 & 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1/2 \end{bmatrix}$$

$$\vec{c} = \begin{bmatrix} 2 \\ 3 \\ 1/2 \end{bmatrix}$$

Solve, given L and U.

- Solve $U\vec{x} = \vec{c}$:

$$U\vec{x} = \vec{c}, U = \begin{bmatrix} 2 & 3 & 4 & 4 \\ 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & -3/2 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c} = \begin{bmatrix} 2 \\ 3 \\ 1/2 \end{bmatrix}$$

$$M = \begin{bmatrix} 2 & 3 & 4 & 4 & 2 \\ 0 & -2 & 0 & -1 & 3 \\ 0 & 0 & 0 & -5/2 & 1/2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 3 & 4 & 4 & 2 \\ 0 & -2 & 0 & -1 & 3 \\ 0 & 0 & 0 & 1 & -1/5 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 3 & 4 & 0 & 14/5 \\ 0 & -2 & 0 & 0 & 14/5 \\ 0 & 0 & 0 & 1 & -1/5 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 & 3 & 4 & 0 & 14/5 \\ 0 & 1 & 0 & 0 & -7/5 \\ 0 & 0 & 0 & 1 & -1/5 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 0 & 4 & 0 & 7 \\ 0 & 1 & 0 & 0 & -7/5 \\ 0 & 0 & 0 & 1 & -1/5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 & 0 & 7/2 \\ 0 & 1 & 0 & 0 & -7/5 \\ 0 & 0 & 0 & 1 & -1/5 \end{bmatrix}$$

Solve, given L and U.

- Solve $U\vec{x} = \vec{c}$:

$$U\vec{x} = \vec{c}, U = \begin{bmatrix} 2 & 3 & 4 & 4 \\ 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & -3/2 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c} = \begin{bmatrix} 2 \\ 3 \\ 1/2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 & 7/2 \\ 0 & 1 & 0 & 0 & -7/5 \\ 0 & 0 & 0 & 1 & -1/5 \end{bmatrix} \rightarrow \begin{array}{rcl} x_1 & + & 2x_3 = 7/2 \\ x_2 & & = -7/5 \\ x_4 & & = -1/5 \end{array}$$

$$\begin{array}{rcl} x_1 & = & 7/2 - 2x_3 \\ x_2 & = & -7/5 \\ x_4 & = & -1/5 \end{array} \rightarrow \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 7/2 - 2t \\ -7/5 \\ t \\ -1/5 \end{bmatrix}, \forall t \in \mathbb{R}$$

Solve, given P, L and U.

- Solve the linear system $A\vec{x} = \vec{b}$, given:

$$PA = LU,$$

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 1/2 & 0 & 0 & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} 1 \\ 0 \\ 2 \\ -1 \end{bmatrix}$$

Solve, given P, L and U.

- Apply row swaps to RHS:

$$A\vec{x} = \vec{b}, PA = LU \rightarrow L(U\vec{x}) = P\vec{b} \rightarrow \vec{b}' = P\vec{b}, L\vec{c} = \vec{b}', U\vec{x} = \vec{c}$$

$$\vec{b}' = P\vec{b} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 2 \end{bmatrix}$$

Solve, given P, L and U.

- Set up two triangular systems:

$$L\vec{c} = \vec{b}', L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 1/2 & 0 & 0 & 1 \end{bmatrix}, \vec{c} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}, \vec{b}' = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 2 \end{bmatrix}$$

$$U\vec{x} = \vec{c}, U = \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}$$

Solve, given P, L and U.

- Solve $L\vec{c} = \vec{b}'$:

$$L\vec{c} = \vec{b}', L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 1/2 & 0 & 0 & 1 \end{bmatrix}, \vec{c} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}, \vec{b}' = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 2 \end{bmatrix}$$

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & -2 & 1 & 0 & -1 \\ 1/2 & 0 & 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & -2 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

$$\vec{c} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix}$$

Solve, given P, L and U.

- Solve $U\vec{x} = \vec{c}$:

$$U\vec{x} = \vec{c}, U = \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix}$$

$$M = \begin{bmatrix} 2 & 2 & 4 & 4 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & -2 & -2 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 & 4 & 0 & -8 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & -2 & 0 & 5 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 & 4 & 0 & -8 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -5/2 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 & 2 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -5/2 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 0 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -5/2 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -5/2 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

Solve, given P, L and U.

- Solve $U\vec{x} = \vec{c}$:

$$U\vec{x} = \vec{c}, U = \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -5/2 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} \rightarrow \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ -5/2 \\ 2 \end{bmatrix}$$

Solve a system with multiple RHS.

- Solve the given linear system using $Lc=b$ and $Ux=c$ for two different right-hand sides.

$$\begin{array}{cccccccl} 2x_1 & & & - & 2x_3 & + & 4x_4 & = & d_1 \\ 2x_1 & + & 2x_2 & + & 2x_3 & + & 6x_4 & = & d_2 \\ x_1 & - & x_2 & - & 3x_3 & & & = & d_3 \\ & & 3x_2 & + & 6x_3 & + & 3x_4 & = & d_4 \end{array}$$

Case 1: $(d_1, d_2, d_3, d_4) = (2, 2, 1, 5)$

Case 2: $(d_1, d_2, d_3, d_4) = (2, 4, 2, 3)$

Solve a system with multiple RHS.

- Express in matrix form:

$$\begin{array}{cccccccl} 2x_1 & & & - & 2x_3 & + & 4x_4 & = & d_1 \\ 2x_1 & + & 2x_2 & + & 2x_3 & + & 6x_4 & = & d_2 \\ x_1 & - & x_2 & - & 3x_3 & & & = & d_3 \\ & & 3x_2 & + & 6x_3 & + & 3x_4 & = & d_4 \end{array}$$

Case 1: $(d_1, d_2, d_3, d_4) = (2, 2, 1, 5)$

Case 2: $(d_1, d_2, d_3, d_4) = (2, 4, 2, 3)$

$$A\vec{x} = \vec{b}, \quad A = \begin{bmatrix} 2 & 0 & -2 & 4 \\ 2 & 2 & 2 & 6 \\ 1 & -1 & -3 & 0 \\ 0 & 3 & 6 & 3 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{b}_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \\ 5 \end{bmatrix}, \vec{b}_2 = \begin{bmatrix} 2 \\ 4 \\ 2 \\ 3 \end{bmatrix}$$

Solve a system with multiple RHS.

- Perform Forward Elimination to get U and encode L:

$$A = \begin{bmatrix} 2 & 0 & -2 & 4 \\ 2 & 2 & 2 & 6 \\ 1 & -1 & -3 & 0 \\ 0 & 3 & 6 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 0 & -2 & 4 \\ 0 & 2 & 4 & 2 \\ 0 & -1 & -2 & -2 \\ 0 & 3 & 6 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 0 & -2 & 4 \\ 0 & 2 & 4 & 2 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 0 & -2 & 4 \\ 0 & 2 & 4 & 2 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & 0 \\ & & 1 & 0 \\ & & & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & 0 \\ 1/2 & & 1 & 0 \\ 0 & & & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & 0 \\ 1/2 & -1/2 & 1 & 0 \\ 0 & 3/2 & & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & 0 \\ 1/2 & -1/2 & 1 & 0 \\ 0 & 3/2 & 0 & 1 \end{bmatrix}$$

Solve a system with multiple RHS.

- Set up two triangular systems:

$$A\vec{x} = \vec{b}_i, A = LU \rightarrow L(U\vec{x}) = \vec{b}_i \rightarrow L\vec{c}_i = \vec{b}_i, U\vec{x} = \vec{c}_i$$

$$L\vec{c} = \vec{b}, \quad L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1/2 & -1/2 & 1 & 0 \\ 0 & 3/2 & 0 & 1 \end{bmatrix}, \vec{c}_i = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}, \vec{b}_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \\ 5 \end{bmatrix}, \vec{b}_2 = \begin{bmatrix} 2 \\ 4 \\ 2 \\ 3 \end{bmatrix}$$

$$U\vec{x} = \vec{c}, \quad U = \begin{bmatrix} 2 & 0 & -2 & 4 \\ 0 & 2 & 4 & 2 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c}_i = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}$$

Solve a system with multiple RHS.

- Solve $L\vec{c}_1 = \vec{b}_1$:

$$L\vec{c}_1 = \vec{b}_1, L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1/2 & -1/2 & 1 & 0 \\ 0 & 3/2 & 0 & 1 \end{bmatrix}, \vec{c}_1 = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}, \vec{b}_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \\ 5 \end{bmatrix}$$

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1/2 & -1/2 & 1 & 0 & 1 \\ 0 & 3/2 & 0 & 1 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & -1/2 & 1 & 0 & 0 \\ 0 & 3/2 & 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

$$\vec{c}_1 = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 2 \end{bmatrix}$$

Solve a system with multiple RHS.

- Solve $U\vec{x} = \vec{c}_1$:

$$U\vec{x} = \vec{c}_1, U = \begin{bmatrix} 2 & 0 & -2 & 4 \\ 0 & 2 & 4 & 2 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c}_1 = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 2 \end{bmatrix}$$

$$M = \begin{bmatrix} 2 & 0 & -2 & 4 & 2 \\ 0 & 2 & 4 & 2 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$$

- System is inconsistent, there is no solution for this right-hand side.

Solve a system with multiple RHS.

- Solve $L\vec{c}_2 = \vec{b}_2$:

$$L\vec{c}_2 = \vec{b}_2, L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1/2 & -1/2 & 1 & 0 \\ 0 & 3/2 & 0 & 1 \end{bmatrix}, \vec{c}_2 = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}, \vec{b}_2 = \begin{bmatrix} 2 \\ 4 \\ 2 \\ 3 \end{bmatrix}$$

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 1 & 1 & 0 & 0 & 4 \\ 1/2 & -1/2 & 1 & 0 & 2 \\ 0 & 3/2 & 0 & 1 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & -1/2 & 1 & 0 & 1 \\ 0 & 3/2 & 0 & 1 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\vec{c}_2 = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 0 \end{bmatrix}$$

Solve a system with multiple RHS.

- Solve $U\vec{x} = \vec{c}_2$:

$$U\vec{x} = \vec{c}_2, U = \begin{bmatrix} 2 & 0 & -2 & 4 \\ 0 & 2 & 4 & 2 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c}_2 = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 0 \end{bmatrix}$$

$$M = \begin{bmatrix} 2 & 0 & -2 & 4 & 2 \\ 0 & 2 & 4 & 2 & 2 \\ 0 & 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 0 & -2 & 4 & 2 \\ 0 & 2 & 4 & 2 & 2 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 0 & -2 & 0 & 10 \\ 0 & 2 & 4 & 0 & 6 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 & 0 & -2 & 0 & 10 \\ 0 & 1 & 2 & 0 & 3 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 & 0 & 5 \\ 0 & 1 & 2 & 0 & 3 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Solve a system with multiple RHS.

- Solve $U\vec{x} = \vec{c}_2$:

$$U\vec{x} = \vec{c}_2, U = \begin{bmatrix} 2 & 0 & -2 & 4 \\ 0 & 2 & 4 & 2 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c}_2 = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -1 & 0 & 5 \\ 0 & 1 & 2 & 0 & 3 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{array}{rclcl} x_1 & - & x_3 & = & 5 \\ x_2 & + & 2x_3 & = & 3 \\ & & x_4 & = & -2 \end{array}$$

$$\begin{array}{rcl} x_1 & = & 5 + x_3 \\ x_2 & = & 3 - 2x_3 \\ x_4 & = & -2 \end{array} \rightarrow \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 5 + t \\ 3 - 2t \\ t \\ -2 \end{bmatrix}, \forall t \in \mathbb{R}$$

Solve a system with multiple RHS.

- Solve the given linear system using $Lc=b$ and $Ux=c$ for two different right-hand sides.

$$\begin{array}{cccccccl} 2x_1 & & & - & 2x_3 & + & 4x_4 & = & d_1 \\ 2x_1 & + & 2x_2 & + & 2x_3 & + & 6x_4 & = & d_2 \\ x_1 & - & x_2 & - & 3x_3 & & & = & d_3 \\ & & 3x_2 & + & 6x_3 & + & 3x_4 & = & d_4 \end{array}$$

$(d_1, d_2, d_3, d_4) = (2, 2, 1, 5) \rightarrow$ **No solution.**

$(d_1, d_2, d_3, d_4) = (2, 4, 2, 3) \rightarrow (x_1, x_2, x_3, x_4) = (5 + t, 3 - 2t, t, -2), \forall t \in \mathbb{R}$

Unit 5.4

The LDU decomposition.

LDU decomposition.

- As derived, L has ones on the diagonal, but U may have any diagonal values.
- If we add in only the pivot scaling step of back substitution, U will also have ones at all pivot positions.

$$A \xrightarrow{\text{Row Exchanges}} A' \xrightarrow{\text{Elimination}} U' \xrightarrow{\text{Pivot Scaling}} U$$

- The pivot scaling step can be represented by multiplication by a diagonal matrix.

LDU decomposition.

- The scaling matrix contains the inverse of the diagonal entries:

$$D^{-1}U' = U$$

$$\begin{bmatrix} 1/u_{11} & 0 & 0 & \cdots & 0 \\ 0 & 1/u_{22} & 0 & \cdots & 0 \\ 0 & 0 & 1/u_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \cdots & 1/u_{mm} \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} & \cdots & u_{1n} \\ 0 & u_{22} & u_{23} & \cdots & u_{2n} \\ 0 & 0 & u_{33} & \cdots & u_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \cdots & u_{mn} \end{bmatrix} = \begin{bmatrix} 1 & u_{12}/u_{11} & u_{13}/u_{11} & \cdots & u_{1n}/u_{11} \\ 0 & 1 & u_{23}/u_{22} & \cdots & u_{2n}/u_{22} \\ 0 & 0 & 1 & \cdots & u_{3n}/u_{33} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \cdots & 1 \end{bmatrix}$$

LDU decomposition.

- We can also swap the order, using a diagonal matrix to “undo” the scaling.
- In this form, we have a diagonal matrix that simply contains the diagonal entries themselves:

$$DU = U'$$

$$\begin{bmatrix} u_{11} & 0 & 0 & \cdots & 0 \\ 0 & u_{22} & 0 & \cdots & 0 \\ 0 & 0 & u_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \cdots & u_{mm} \end{bmatrix} \begin{bmatrix} 1 & u_{12}/u_{11} & u_{13}/u_{11} & \cdots & u_{1n}/u_{11} \\ 0 & 1 & u_{23}/u_{22} & \cdots & u_{2n}/u_{22} \\ 0 & 0 & 1 & \cdots & u_{3n}/u_{33} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \cdots & 1 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} & \cdots & u_{1n} \\ 0 & u_{22} & u_{23} & \cdots & u_{2n} \\ 0 & 0 & u_{33} & \cdots & u_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \cdots & u_{mn} \end{bmatrix}$$

LDU decomposition.

$$PA = LDU$$
$$A \xrightarrow{\text{Row Exchanges}} A' \xrightarrow{\text{Elimination}} U' \xrightarrow{\text{Pivot Scaling}} U$$
$$A' = PA \quad A' = LU' \quad U' = DU$$

- L and U are triangular (lower and upper) with ones on the diagonal.
- A and U are m -by- n matrices; P, L and D are m -by- m matrices.
- P and L are always non-singular.
- D non-singular if (and only if) no zero rows appear during forward elimination; D non-singular if and only if every RHS has a solution.

LDU with singular matrices.

- If U contains zero rows or free variables, D^{-1} often does not exist.

$$D^{-1}U' = U$$

$$\begin{bmatrix} 1/u_{11} & 0 \\ 0 & \mathbf{1/0} \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} & u_{14} \\ 0 & 0 & u_{23} & u_{24} \end{bmatrix} = \begin{bmatrix} 1 & u_{12}/u_{11} & u_{13}/u_{11} & u_{14}/u_{11} \\ ? & ? & ? & ? \end{bmatrix}$$

$$\begin{bmatrix} 1/u_{11} & 0 & 0 & 0 \\ 0 & 1/u_{22} & 0 & 0 \\ 0 & 0 & 1/u_{33} & 0 \\ 0 & 0 & 0 & \mathbf{1/0} \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & u_{12}/u_{11} & u_{13}/u_{11} \\ 0 & 1 & u_{23}/u_{22} \\ 0 & 0 & 1 \\ ? & ? & ? \end{bmatrix}$$

LDU with singular matrices.

- In some cases we can still define D , but with ambiguity and/or deviation from the standard.

$$DU = U'$$

$$\begin{bmatrix} u_{11} & 0 \\ 0 & \mathbf{1} \end{bmatrix} \begin{bmatrix} 1 & u_{12}/u_{11} & u_{13}/u_{11} & u_{14}/u_{11} \\ 0 & \mathbf{0} & u_{23} & u_{24} \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} & u_{14} \\ 0 & 0 & u_{23} & u_{24} \end{bmatrix}$$

$$\begin{bmatrix} u_{11} & 0 & 0 & 0 \\ 0 & u_{22} & 0 & 0 \\ 0 & 0 & u_{33} & 0 \\ 0 & 0 & 0 & \mathbf{0} \end{bmatrix} \begin{bmatrix} 1 & u_{12}/u_{11} & u_{13}/u_{11} \\ 0 & 1 & u_{23}/u_{22} \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \\ 0 & 0 & 0 \end{bmatrix}$$

LDU with singular matrices.

- In square matrices, we can always define D , with no deviation from the standard.
- To achieve this we don't move zero rows to the bottom; we move them to row(s) that matches the free variable(s)!

$$DU = U'$$

$$\begin{bmatrix} u_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & u_{33} & 0 \\ 0 & 0 & 0 & u_{44} \end{bmatrix} \begin{bmatrix} 1 & u_{12}/u_{11} & u_{13}/u_{11} & u_{14}/u_{11} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & u_{34}/u_{33} \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} & u_{14} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & u_{33} & u_{34} \\ 0 & 0 & 0 & u_{44} \end{bmatrix}$$